

Example 4.54. For $m < n \leq k$ there are fiber bundles

$$V_{n-m}(\mathbb{R}^{k-m}) \rightarrow V_n(\mathbb{R}^k) \xrightarrow{p} V_m(\mathbb{R}^k)$$

where the projection p sends an n -frame onto the m -frame formed by its first m vectors, so the fiber consists of $(n-m)$ -frames in the $(k-m)$ -plane orthogonal to a given m -frame. Local trivializations can be constructed as follows. For an m -frame F , choose an orthonormal basis for the $(k-m)$ -plane orthogonal to F . This determines orthonormal bases for the $(k-m)$ -planes orthogonal to all nearby m -frames by orthogonal projection and Gram-Schmidt, as in the preceding example. This allows us to identify these $(k-m)$ -planes with $\mathbb{R}^{k-m}$, and in particular the fibers near $p^{-1}(F)$ are identified with $V_{n-m}(\mathbb{R}^{k-m})$, giving a local trivialization.

There are analogous bundles in the complex and quaternionic cases as well, with local triviality shown in the same way.

Restricting to the case $m = 1$, we have bundles $V_{n-1}(\mathbb{R}^{k-1}) \rightarrow V_n(\mathbb{R}^k) \rightarrow S^{k-1}$ whose associated long exact sequence of homotopy groups allows us deduce that $V_n(\mathbb{R}^k)$ is $(k-n-1)$ -connected by induction on n . In the complex and quaternionic cases the same argument yields the other connectivity statements in the preceding example.

Taking $k = n$ we obtain fiber bundles $O(k-m) \rightarrow O(k) \rightarrow V_m(\mathbb{R}^k)$. The fibers are in fact just the cosets $\alpha O(k-m)$ for $\alpha \in O(k)$, where $O(k-m)$ is regarded as the subgroup of $O(k)$ fixing the first m standard basis vectors. So we see that $V_m(\mathbb{R}^k)$ is identifiable with the coset space $O(k)/O(k-m)$, or in other words the orbit space for the free action of $O(k-m)$ on $O(k)$ by right-multiplication. In similar fashion one can see that $G_m(\mathbb{R}^k)$ is the coset space $O(k)/(O(m) \times O(k-m))$ where the subgroup $O(m) \times O(k-m) \subset O(k)$ consists of the orthogonal transformations taking the m -plane spanned by the first m standard basis vectors to itself. The corresponding observations apply also in the complex and quaternionic cases, with the unitary and symplectic groups.

Example 4.55: Bott Periodicity. Specializing the preceding example by taking $m = 1$ and $k = n$ we obtain bundles

$$\begin{aligned} O(n-1) &\rightarrow O(n) \xrightarrow{p} S^{n-1} \\ U(n-1) &\rightarrow U(n) \xrightarrow{p} S^{2n-1} \\ Sp(n-1) &\rightarrow Sp(n) \xrightarrow{p} S^{4n-1} \end{aligned}$$

The map p can be described as evaluation of an orthogonal, unitary, or symplectic transformation on a fixed unit vector. These bundles show that computing homotopy groups of $O(n)$, $U(n)$, and $Sp(n)$ should be at least as difficult as computing homotopy groups of spheres. For example, if one knew the homotopy groups of $O(n)$ and $O(n-1)$, then from the long exact sequence of homotopy groups for the first bundle one could say quite a bit about the homotopy groups of S^{n-1} .

The bundles above imply a very interesting stability property. In the real case, the inclusion $O(n-1) \hookrightarrow O(n)$ induces an isomorphism on π_i for $i < n-2$, from the long exact sequence of the first bundle. Hence the groups $\pi_i O(n)$ are independent of n if n is sufficiently large, and the same is true for the groups $\pi_i U(n)$ and $\pi_i Sp(n)$ via the other two bundles. One of the most surprising results in all of algebraic topology is the Bott Periodicity Theorem which asserts that these stable groups repeat periodically, with a period of eight for O and Sp and a period of two for U . Their values are given in the following table:

$i \bmod 8$	0	1	2	3	4	5	6	7
$\pi_i O(n)$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$
$\pi_i U(n)$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$\pi_i Sp(n)$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$

Stable Homotopy Groups

We showed in Corollary 4.24 that for an n -connected CW complex X , the suspension map $\pi_i(X) \rightarrow \pi_{i+1}(SX)$ is an isomorphism for $i < 2n+1$. In particular this holds for $i \leq n$ so SX is $(n+1)$ -connected. This implies that in the sequence of iterated suspensions

$$\pi_i(X) \rightarrow \pi_{i+1}(SX) \rightarrow \pi_{i+2}(S^2X) \rightarrow \cdots$$

all maps are eventually isomorphisms, even without any connectivity assumption on X itself. The resulting **stable homotopy group** is denoted $\pi_i^s(X)$.

An especially interesting case is the group $\pi_i^s(S^0)$, which equals $\pi_{i+n}(S^n)$ for $n > i+1$. This stable homotopy group is often abbreviated to π_i^s and called the **stable i -stem**. It is a theorem of Serre which we prove in [SSAT] that π_i^s is always finite for $i > 0$.

These stable homotopy groups of spheres are among the most fundamental objects in topology, and much effort has gone into their calculation. At the present time, complete calculations are known only for i up to around 60 or so. Here is a table for $i \leq 19$, taken from [Toda 1962]:

i	0	1	2	3	4	5	6	7	8	9	10	11	12
π_i^s	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{240}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_6$	$\mathbb{Z}_{504}$	0
	13	14			15		16		17		18		19
	$\mathbb{Z}_3$	$\mathbb{Z}_2 \times \mathbb{Z}_2$			$\mathbb{Z}_{480} \times \mathbb{Z}_2$		$\mathbb{Z}_2 \times \mathbb{Z}_2$		$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$		$\mathbb{Z}_8 \times \mathbb{Z}_2$		$\mathbb{Z}_{264} \times \mathbb{Z}_2$

Patterns in this apparent chaos begin to emerge only when one projects π_i^s onto its **p -components**, the quotient groups obtained by factoring out all elements of order relatively prime to the prime p . For $i > 0$ the p -component ${}_p\pi_i^s$ is of course isomorphic to the subgroup of π_i^s consisting of elements of order a power of p , but the quotient viewpoint is in some ways preferable.